

Gosford High School

Year 11

2008

Preliminary Higher School Certificate

Mathematics

Assessment Task 1

Time Allowed - 60 minutes

Remember to start each new question on a new page

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

Question 1 (20 Marks)

- (a) Write 32 as a percentage of 56 (correct to the nearest percent) (1)
- (b) The population of the world is approximately 6700 million (or 6.7 billion) people. Write this population using scientific notation. (1)
- (c) Find $\sqrt[5]{6.3 \times 10^6}$ correct to the nearest whole number (2)
- (d) Evaluate $\sqrt{\frac{3^2 + 4^2}{12^2 + 5^2}}$ (1)
- (e) $-12, \sqrt{-6}, \frac{3}{7}, 2.\dot{5}, 0, \sqrt{8}, 12.5, 33\frac{1}{3}\%, \pi, |14|, 7^{-3}$
From the above set of numerals list the set of
- (i) Rational Numbers (2)
- (ii) Irrational Numbers (1)
- (f) Write $0.5\dot{1}\dot{8}$ as a simplified proper fraction (3)
- (g) Write $\frac{-5 + \sqrt{17}}{8}$ correct to 2 decimal places (1)
- (h) A store's sale price advertises 15% off the marked price of all suits. If Angus pays \$153 for his suit what was the marked price of the suit. (2)
- (i) Write (i) 0.00316 correct to 1 significant figure (1)
- (ii) 108964091 correct to 4 significant figures (1)
- (j) Simplify $\sqrt{(-4)^2} - |-3| \times |9 - 4|$ (1)
- (k) Gold, a very soft metal, can be hammered into flat sheets of thickness 1.02×10^{-4} mm. How many sheets are needed to make a pile 1 cm thick? (answer to the nearest sheet) (2)
- (l) Evaluate $\frac{3.6 \times 10^{16}}{2.4 \times 10^{11} \times 1.2 \times 10^9}$, writing your answer as a decimal (1)

Question 2 (15 Marks)

- (a) $\sqrt{60}$ lies between which two consecutive integers? (1)
- (b) Find the value of m if $8\sqrt{2} = \sqrt{m}$ (1)
- (c) Simplify $\frac{5\sqrt{294}}{35\sqrt{3}}$ (2)
- (d) Simplify $4\sqrt{32} \times 5\sqrt{75}$ (2)
- (e) Simplify $\sqrt{200} - 6\sqrt{18}$ (2)
- (f) Find x and y if $(2\sqrt{3} - 3\sqrt{2})^2 = x + y\sqrt{6}$ (2)
- (g) Express $\frac{\sqrt{3} + 2}{3\sqrt{3} - 2}$ with a rational denominator (2)
- (h) Triangle ABC is right angled at C, $a = \sqrt{5} - 2$, $b = \sqrt{5} + 2$.
Find c in simplest surd form. (3)

Question 3 (15 Marks)

- (a) Find all possible values of k if $16x^2 + kx + 25$ is a perfect square (2)
- (b) Factorise each of the following
- (i) $18y^2 - 50$ (2)
- (ii) $3x^2 - 3xy + 5y - 5x$ (1)
- (iii) $x^3 - 27$ (1)
- (iv) $9k^2 - 42k + 49$ (1)
- (v) $10x^2 + x - 21$ (1)
- (vi) $y^4 - y^2 - 12$ (2)
- (c) Expand and simplify
- (i) $(2x + 3y)(2x - 3y)$ (1)
- (ii) $(e^x - 2)^2$ (1)
- (iii) $(3x - 1)^2 - 2(3x + 1)(3x - 1)$ (3)

Question 4 (20 Marks)

(a) Subtract $4x - 5y + 7$ from $3x - 2y$ (1)

(b) Find y when $x = -3$ given $y = 1 - 5x - 2x^3$ (1)

(c) Given $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$ find the exact value of d (in simplest form) given
 $a = 3, b = -2, c = -4, x_1 = 6, y_1 = 20$ (3)

(d) Simplify $(16x^{-2})^{-\frac{3}{2}}$ express your answer without fractional or negative indices (2)

(e) Solve $27^x = \frac{1}{9}$ (1)

(f) Simplify $\frac{4^{-1}x^{\frac{5}{2}}}{2^{-3}x^{\frac{1}{2}}}$ (1)

(g) Simplify $\frac{x^{-1} + y^{-1}}{x + y}$ (2)

(h) Simplify

(i) $\frac{2a-3}{6} - \frac{3a+1}{9}$ (2)

(ii) $\frac{a^2m - 4m^3}{6a^2m} \div \frac{a^2 + am - 6m^2}{3a^2 + 9am}$ (3)

(iii) $\frac{2}{x^2-1} - \frac{1}{x^2-x} + \frac{x-1}{x^2+x}$ (4)

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- 1) a) 57%
 b) 6.7×10
 c) 23
 d) $5/13$
 e) i) $-12, \frac{3}{7}, 2.5, 0, 12.5, 33\frac{1}{3}\%, 14, 7^3$
 ii) $\sqrt{-6}, \sqrt{8}, \pi$
 f) let $x = 0.51818...$

$$100x = 51.81818...$$

$$99x = 51.3$$

$$x = \frac{513}{990}$$

$$= 51/110$$

g) -0.11

h) 85% of marked price = 153
 marked price = \$180

i) i) 0.003

ii) 109000 000

j) -11

k) $10 \div (1.02 \times 10^4) = 98039$

l) 1.25×10^{-4}

2) a) 7 and 8

b) $m = 128$

c) $\frac{\sqrt{98}}{7} = \frac{7\sqrt{2}}{7}$

$$= \sqrt{2}$$

d) $16\sqrt{2} \times 25\sqrt{3} = 400\sqrt{6}$

e) $10\sqrt{2} - 18\sqrt{2} = -8\sqrt{2}$

f) $12 - 12\sqrt{6} + 18 = 30 - 12\sqrt{6}$

$$x = 30 \text{ and } y = -12$$

g) $\frac{\sqrt{3}+2}{3\sqrt{3}-2} \times \frac{3\sqrt{3}+2}{3\sqrt{3}+2} = \frac{9+8\sqrt{3}+4}{27-4}$

$$= \frac{13+8\sqrt{3}}{23}$$

h) $c^2 = (\sqrt{5}-2)^2 + (\sqrt{5}+2)^2$
 $= 9 - 4\sqrt{5} + 9 + 4\sqrt{5}$
 $= 18$

$$c = 3\sqrt{2}$$

3) a) $(4x \pm 5)^2$
 $k = \pm 20$

b) i) $2(9y^2 - 25)$
 $= 2(3y-5)(3y+5)$

ii) $3x(x-y) - 5(x-y)$
 $= (x-y)(3x-5)$

iii) $(x-3)(x+3x+9)$

iv) $(3k-7)^2$

v) $(5x-7)(2x+3)$

vi) $(y^2-4)(y^2+3)$
 $= (y+2)(y-2)(y^2+3)$

c) i) $4x^2 - 9$

ii) $e^2 - 4e + 4$

iii) $9x^2 - 6x + 1 - 2(9x^2 - 1)$
 $= 9x^2 - 6x + 1 - 18x^2 + 2$
 $= -9x^2 - 6x + 3$

4) a) $3x - 2y - (4x - 5y + 7)$
 $= 3x - 2y - 4x + 5y - 7$
 $= -x + 3y - 7$

b) $y = 1 + 15 - 18$
 $= -2$

c) $d = \frac{18 + -40 - 4}{\sqrt{3^2 + (-2)^2}}$
 $= \frac{-26}{\sqrt{13}}$

d) $(16x^{-2})^{\frac{3}{2}} = \left(\sqrt{\frac{16}{x^2}}\right)^3$
 $= \frac{64}{x^3}$

e) $3^{3x} = 3^{-2}$

f) $\frac{x^{\frac{3}{2}}}{4} \div \frac{1}{8x^{\frac{1}{2}}} = 2x^3$

g) $\frac{1}{x} + \frac{1}{y} \div (x+y)$
 $= \frac{x+y}{xy} \times \frac{1}{x+y}$
 $= \frac{1}{xy}$

$$\begin{aligned}
 \text{h) i)} \quad & \frac{2a-3}{6} - \frac{3a+1}{9} \\
 & = \frac{6a-9-2(3a+1)}{18} \\
 & = \frac{6a-9-6a-2}{18} \\
 & = \frac{-11}{18}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad & \frac{x(a^2-4m^2)}{6a^2m} \times \frac{3x(a+m)}{(a-2m)(a+3m)} \\
 & = \frac{(a+2m)(a-2m)(a+m)}{2a(a-2m)(a+3m)} \\
 & = \frac{(a+2m)(a+m)}{2a(a+3m)}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii)} \quad & \frac{2}{(x+1)(x-1)} - \frac{1}{x(x-1)} + \frac{x-1}{x(x+1)} \\
 & = \frac{2x-1(x+1)+(x-1)(x-1)}{x(x+1)(x-1)} \\
 & = \frac{2x-x-1+x^2-2x+1}{x(x+1)(x-1)} \\
 & = \frac{x(x-1)}{x(x+1)(x-1)} \\
 & = \frac{1}{x+1}
 \end{aligned}$$